Esercitazione 7

Computational Chemical Kinetics and Transition State Theory (TST)

Theoretical Background

The Potential Energy Surface and the Saddle Point

For a gas-phase elementary reaction converting reactants, denoted by R, into products, denoted by P, the nuclear geometry changes along a multidimensional potential energy surface (PES),

$$E = E(\mathbf{R}_1, \mathbf{R}_2, \ldots, \mathbf{R}_N), \tag{1}$$

where E is the electronic energy, N is the number of atoms, and $\mathbf{R}_i$ is the Cartesian position vector of atom i. The minimum-energy path (MEP), also called the intrinsic reaction path when mass-weighted coordinates are used, connects the reactant well to the product well through the lowest-energy route.

The transition state (TS), or activated complex, is denoted by ‡. It is the highest-energy point along the MEP and is topologically a first-order saddle point on the PES. At a stationary point, the gradient vanishes:

$$\nabla E = \mathbf{0}. \tag{2}$$

The Hessian matrix is the matrix of second derivatives,

$$H_{ij} = \frac{\partial^2 E}{\partial q_i \, \partial q_j}, \tag{3}$$

where q_i and q_j are mass-weighted nuclear coordinates. A true transition state for an elementary reaction has exactly one negative Hessian eigenvalue. This negative curvature gives one imaginary vibrational frequency, conventionally written as $\nu^{\ddagger}$ or $\tilde{\nu}^{\ddagger}$ when reported as a wavenumber. Gaussian typically prints this mode as a negative frequency in cm^{-1}; in TST and tunneling formulas, its magnitude is used:

$$\left|\nu^{\ddagger}\right| = c \, \left|\tilde{\nu}^{\ddagger}\right|, \tag{4}$$

where c is the speed of light. The imaginary normal mode is not a stable vibration. It corresponds to motion along the reaction coordinate, i.e., the collective bond-breaking and bond-forming displacement that carries the system from reactants to products.

A frequency calculation is therefore a necessary diagnostic:

- a stable reactant or product minimum must have zero imaginary frequencies;
- a first-order transition state must have exactly one imaginary frequency;
- the imaginary mode must visually correspond to the intended chemical reaction;
- an intrinsic reaction coordinate (IRC) calculation is strongly recommended to verify that the TS connects the intended reactants and products.

0.1 TST Postulate and the Eyring Equation

TST assumes that the reactants are in rapid quasi-equilibrium with the activated complex and that each activated complex crossing the dividing surface toward products leads to reaction. If recrossing occurs, or if quantum tunneling is important, the classical TST rate is modified by a transmission coefficient $\kappa(T)$.

For an elementary gas-phase step with molecularity m, the concentration-standard-state Eyring equation can be written as

$$k(T) = \kappa(T)\,\frac{k_{\mathrm{B}}T}{h}\,(c^{\circ})^{1-m}\exp\left[-\frac{\Delta G_c^{\ddagger,\circ}(T)}{RT}\right]. \tag{5}$$

The same equation can be written in enthalpy–entropy form by using

$$\Delta G_c^{\ddagger,\circ}(T) = \Delta H_c^{\ddagger,\circ}(T) - T\Delta S_c^{\ddagger,\circ}(T), \tag{6}$$

which gives

$$k(T) = \kappa(T)\,\frac{k_{\mathrm{B}}T}{h}\,(c^{\circ})^{1-m}\exp\left(\frac{\Delta S_c^{\ddagger,\circ}}{R}\right)\exp\left(-\frac{\Delta H_c^{\ddagger,\circ}}{RT}\right). \tag{7}$$

0.2 Definitions of the Main Symbols

$k(T)$	Temperature-dependent macroscopic rate constant. Its units depend on reaction order: s^{-1} for a unimolecular elementary step and $\mathrm{L\,mol^{-1}\,s^{-1}}$ or $\mathrm{M^{-1}\,s^{-1}}$ for a bimolecular elementary step when concentration is expressed in molarity.
T	Absolute temperature in kelvin, K.
R	Molar gas constant, $R = 8.314\,462\,618\,\mathrm{J\,mol^{-1}\,K^{-1}}$.
k_{B}	Boltzmann constant, $k_{\mathrm{B}} = 1.380\,649 \times 10^{-23}\,\mathrm{J\,K^{-1}}$.
h	Planck constant, $h = 6.626\,070\,15 \times 10^{-34}\,\mathrm{J\,s}$.
$\kappa(T)$	Transmission coefficient. Classical TST assumes $\kappa = 1$. Tunneling corrections usually give $\kappa > 1$. Dynamical recrossing corrections may give an effective factor smaller than one, but this is not captured by simple Wigner or Eckart tunneling corrections.
m	Molecularity of the elementary step: $m = 1$ for unimolecular reactions and $m = 2$ for bimolecular reactions.
c°	Standard concentration, normally $1\,\mathrm{mol\,L^{-1}} = 1\,\mathrm{M}$ for concentration-based kinetics.
$\Delta G_c^{\ddagger,\circ}$	Standard Gibbs free energy of activation at the concentration standard state c°. For an elementary step, it is the free-energy difference between the transition state and the separated reactants at the same standard state.
$\Delta H_c^{\ddagger,\circ}$	Standard enthalpy of activation at the concentration standard state.
$\Delta S_c^{\ddagger,\circ}$	Standard entropy of activation at the concentration standard state.

Quantum Mechanical Tunneling

Classically, reactants with insufficient energy to cross the activation barrier cannot react. Quantum mechanically, light particles, especially hydrogen atoms and protons, can have a nonzero probability of passing through the barrier. This is quantum tunneling. Tunneling is most important for narrow barriers, light-transferred atoms, and low temperatures.

When tunneling is included through a multiplicative correction,

$$k_{\text{tunnel}}(T) = \kappa(T)\, k_{\text{classical}}(T). \tag{8}$$

If the only change between two calculations is the tunneling model, then

$$\frac{k_{\text{tunnel}}}{k_{\text{classical}}} = \kappa(T). \tag{9}$$

Wigner Correction

The Wigner correction is a first-order analytical approximation based on an inverted parabolic barrier. It requires only the magnitude of the imaginary TS frequency:

$$\kappa_{\text{Wigner}} = 1 + \frac{1}{24}\left(\frac{h\left|\nu^{\ddagger}\right|}{k_{\text{B}}T}\right)^{2}. \tag{10}$$

If the imaginary frequency is given as a wavenumber, $\tilde{\nu}^{\ddagger}$ in cm^{-1}, the equivalent form is

$$\kappa_{\text{Wigner}} = 1 + \frac{1}{24}\left(\frac{hc\left|\tilde{\nu}^{\ddagger}\right|}{k_{\text{B}}T}\right)^{2}. \tag{11}$$

The Wigner correction is most reliable when tunneling is small. It can overestimate or become physically unrealistic at very low temperatures or for very sharp barriers.

Eckart Correction

The Eckart correction uses an analytical one-dimensional barrier shape that can be asymmetric. It accounts for the forward barrier, reverse barrier, reaction energy, and imaginary frequency. It is therefore more physically informative than the Wigner approximation for exothermic or endothermic reactions. In practice, an Eckart calculation requires consistent electronic energies and thermochemical data for

- all reactants,
- the transition state,
- all products.

The Eckart model remains a one-dimensional approximation along the reaction coordinate.

Arrhenius Plots and TST Fits

The Arrhenius plot is usually written as

$$\ln k = \ln A - \frac{E_a}{R}\left(\frac{1}{T}\right). \tag{12}$$

If the plot is approximately linear over a chosen temperature interval, linear regression gives

$$\text{slope} = -\frac{E_a}{R}, \qquad \text{intercept} = \ln A. \tag{13}$$

However, TST contains an explicit $\ln T$ contribution through the factor $k_B T/h$, and tunneling makes $\kappa(T)$ temperature dependent. Therefore, the Arrhenius parameters extracted from a wide temperature range should be reported as fitted apparent parameters, not universal constants. Strong upward curvature at low temperature is a common signature of tunneling.

Practical Workflow: Using the Eyringpy Web App

Application platform: https://eyringpy-proof.streamlit.app/

EYRINGPY is used here to automate the extraction of thermochemical quantities from quantum-chemistry output files, apply standard-state corrections, compute tunneling factors, and generate temperature-dependent TST rate constants. The practical session below assumes Gaussian output files, but the same thermodynamic logic applies to other electronic-structure packages if their data are supported and interpreted consistently.

File Preparation and Quality-Control Checklist

Before using the app, run geometry optimization and frequency calculations for every stationary point at the same electronic-structure level of theory, basis set, dispersion treatment, charge, and spin multiplicity appropriate to each species. For Gaussian, this means an `opt freq` calculation for stable minima and a transition-state optimization followed by a frequency calculation for the TS.

You need output files with extensions such as `.log` or `.out` for the following species:

1. **Reactant(s):** each isolated reactant. Each reactant must have zero imaginary frequencies.
2. **Transition state:** the optimized TS geometry. A typical Gaussian route section may include a keyword such as `opt=(ts,calcfc,noeigentest) freq`; however, the final structure must still be checked carefully. The TS must have exactly one imaginary frequency.
3. **Product(s):** required for an Eckart tunneling correction because the model needs the product-side energetics.

Step 1: Uploading the Computational Data

Locate the upload panel in the web interface and provide the following files.

1. **Reactant files.** Upload all reactant Gaussian output files. For a bimolecular reaction $A+B$, upload both isolated reactants. The total reactant thermochemistry is obtained by summing the appropriate thermodynamic quantities of the separated reactants.
2. **Transition-state file.** Upload the TS output file. The program should extract the electronic energy, zero-point energy, thermal corrections, Gibbs free energy, and the imaginary frequency magnitude needed for Wigner or Eckart tunneling.
3. **Product files.** Upload all product files when using the Eckart tunneling model. Product files are also useful for reporting the reaction enthalpy and reaction free energy.

Step 2: Reaction and Phase Settings

1. **Phase.** Select **Gas**. Do not select a solution or implicit-solvent workflow unless the electronic-structure calculations and thermodynamic model were explicitly designed for solution kinetics. For gas-phase reactor modeling, the gas-phase standard-state treatment must be used.
2. **Molecularity.** Select **Unimolecular** or **Bimolecular**. This setting determines the expected rate-constant units and activates the correct standard-state conversion for the activation free energy.

Step 3: Kinetic Parameters

1. **Temperature range.** Use a temperature interval representative of the reactor or chemical environment. For combustion and thermal cracking examples, a useful array is

$$T = 300\,\mathrm{K}, 350\,\mathrm{K}, 400\,\mathrm{K}, \ldots, 1500\,\mathrm{K}. \tag{14}$$

2. **Frequency scaling factor.** Harmonic vibrational frequencies from electronic-structure calculations are often systematically biased. Use a literature scaling factor appropriate to the chosen method and basis set. For example, a value near 0.96 is often used for some B3LYP calculations, but the exact value must be justified from the method-specific literature.

3. **Tunneling correction.** Choose one of the following:

   - **None**: classical TST baseline, $\kappa = 1$;
   - **Wigner**: simple one-dimensional tunneling estimate based only on $\left|\nu^{\ddagger}\right|$;
   - **Eckart**: asymmetric one-dimensional barrier model requiring reactant, TS, and product energetics.

Step 4: Execution and Output Analysis

Click **Calculate**. The output should provide thermodynamic activation parameters, tunneling factors, and rate constants as a function of temperature.

Kinetic Output

For each temperature, record

$$T, \qquad \kappa(T), \qquad k(T), \qquad \ln k(T), \qquad \frac{1}{T}. \tag{15}$$

Check that the units of $k(T)$ are correct:

- unimolecular elementary step: s^{-1};
- bimolecular elementary step: $\mathrm{M}^{-1}\,\mathrm{s}^{-1}$, $\mathrm{L\,mol}^{-1}\,\mathrm{s}^{-1}$, or $\mathrm{cm}^3\,\mathrm{molecule}^{-1}\,\mathrm{s}^{-1}$, depending on the selected output convention.

Arrhenius Analysis

Construct a plot of $\ln k$ against $1/T$. A linear fit over a defined temperature interval gives

$$E_a = -R(\text{slope}), \qquad A = \exp(\text{intercept}). \tag{16}$$

For tunneling-corrected data, avoid fitting one straight line over a very wide range if the plot is visibly curved. Instead, report the temperature interval used for the fit or use the full tabulated $k(T)$ values.

Practical Exercise: Gas-Phase Radical Abstraction

Objective

Combustion, thermal cracking, and atmospheric oxidation mechanisms often involve radical chain reactions. A key elementary propagation step is bimolecular hydrogen-atom abstraction. Because hydrogen is light and the transfer coordinate can be narrow, quantum tunneling can strongly increase the rate constant at low and moderate temperatures.

Reaction Studied

The model reaction is hydrogen abstraction from methane by the hydroxyl radical:

$$\mathrm{CH_4(g) + OH^{\bullet}(g) \xrightarrow{k(T)} CH_3{}^{\bullet}(g) + H_2O\,(g).} \tag{17}$$

The corresponding activated complex can be represented schematically as

$$[\mathrm{H_3C \cdots H \cdots OH}]^{\ddagger}\,, \tag{18}$$

where the transferring hydrogen atom lies between the carbon and oxygen centers along the reaction coordinate.

Computational Setup

Using the provided Gaussian output files, upload the following species:

- methane, CH_4;
- hydroxyl radical, $OH^{\bullet}$;
- transition state, $[H_3C \cdots H \cdots OH]^{\ddagger}$;
- methyl radical, $CH_3{}^{\bullet}$;
- water, H_2O.

Use these settings:

- phase: **Gas**;
- molecularity: **Bimolecular**;
- temperature range: 300 K to 1500 K (Down to 100 K if no appreciable tunneling deviation is found);
- temperature step: 50 K;
- standard concentration: 1 M, if concentration-based rate constants are required;
- tunneling options: first **None**, then **Eckart**.

Classical Versus Quantum Kinetics

Perform two calculations.

1. **Classical TST baseline.** Set tunneling to **None**. Export the full table of T, $k_{\mathrm{classical}}(T)$, $\ln k$, and $1/T$.
2. **Eckart tunneling calculation.** Set tunneling to **Eckart**. Export the full table of T, $\kappa_{\mathrm{Eckart}}(T)$, $k_{\mathrm{Eckart}}(T)$, $\ln k$, and $1/T$.

At each temperature, calculate the tunneling enhancement factor:

$$F_{\mathrm{tun}}(T) = \frac{k_{\mathrm{Eckart}}(T)}{k_{\mathrm{classical}}(T)}. \tag{19}$$

If the two runs are identical except for the tunneling correction, then

$$F_{\mathrm{tun}}(T) = \kappa_{\mathrm{Eckart}}(T). \tag{20}$$

Discussion Questions for the Final Report

Thermodynamic Bottlenecks

Examine $\Delta H^\ddagger$, $\Delta S^\ddagger$, and $\Delta G^\ddagger$. Decide whether the dominant bottleneck is enthalpic, entropic, or both. In this bimolecular gas-phase abstraction, a strongly negative $\Delta S^\ddagger$ is expected because two independently translating and rotating gas-phase species must form one organized activated complex. The entropic penalty is reflected in a reduced pre-exponential factor.

A useful decomposition is

$$\Delta G^\ddagger = \Delta H^\ddagger - T\Delta S^\ddagger. \tag{21}$$

At a given temperature, compare $\Delta H^\ddagger$ with $-T\Delta S^\ddagger$. If $-T\Delta S^\ddagger$ is comparable to or larger than $\Delta H^\ddagger$, the reaction has a major entropic bottleneck.

Rate Enhancement at Room Temperature

At 300 K, calculate

$$\frac{k_{\text{Eckart}}(300\,\text{K})}{k_{\text{classical}}(300\,\text{K})}. \tag{22}$$

This ratio measures how much quantum tunneling accelerates the reaction relative to classical barrier crossing. For a hydrogen-transfer reaction, this factor can be very large at low temperature. Report the numerical value.

Arrhenius Deviations

Plot both data sets, classical and Eckart-corrected, as $\ln k$ versus $1/T$ on the same graph. The classical data are usually closer to linear over moderate temperature ranges. The Eckart-corrected data may curve upward on the right-hand side of the plot, corresponding to low temperature. This curvature occurs because tunneling becomes more important as thermal over-barrier crossing becomes less probable.

At high temperatures, such as 1500 K, many molecules have enough thermal energy to pass over the barrier. The relative importance of tunneling decreases, and $\kappa(T)$ tends toward one. Therefore, the tunneling-corrected rate approaches the classical TST rate in high-temperature combustion conditions.

If you don't observe a curvature, go as low as 100 K.

Additional Exercises

In this exercise, students will study two simple gas-phase unimolecular isomerization reactions using density functional theory and transition state theory:

$$\text{HNC} \longrightarrow \text{HCN}, \tag{23}$$

$$\text{cis-HCOOH} \longrightarrow \text{trans-HCOOH}. \tag{24}$$

The goal is to compute and compare temperature-dependent rate constants using classical transition state theory, the Wigner tunneling correction, and the Eckart tunneling correction. You will also investigate how the predicted activation parameters and kinetic constants depend on the chosen density functional and basis set.

Both reactions are small enough to be treated at several levels of theory within a reasonable computational time. At the same time, they are chemically meaningful examples of unimolecular rearrangement. The first reaction, HNC $\rightarrow$ HCN, involves the migration of a hydrogen atom and is therefore a useful model for discussing quantum tunneling. The second reaction, cis-HCOOH $\rightarrow$ trans-HCOOH, is a conformational isomerization involving internal rotation

around the C–O bond and provides a useful comparison with a heavier-atom rearrangement coordinate.

Reactions Studied

Reaction 1: HNC to HCN Isomerization

The first reaction is

$$\mathrm{HNC}(g) \longrightarrow \mathrm{HCN}(g). \tag{25}$$

The reactant is hydrogen isocyanide, HNC, and the product is hydrogen cyanide, HCN. The transition state corresponds approximately to a hydrogen-transfer geometry connecting the N-bound and C-bound hydrogen structures:

$$[\mathrm{H}\cdots\mathrm{CN}]^{\ddagger}. \tag{26}$$

Because the transferred atom is hydrogen, this reaction is expected to show a visible tunneling effect, especially at low temperature.

Reaction 2: cis-Formic Acid to trans-Formic Acid Isomerization

The second reaction is

$$\text{cis-HCOOH}(g) \longrightarrow \text{trans-HCOOH}(g). \tag{27}$$

This is a unimolecular conformational isomerization. The transition state corresponds primarily to torsion around the C–O bond, passing through a geometry in which the O–H bond rotates from the cis arrangement to the trans arrangement. Since the reaction coordinate is less purely hydrogen-transfer-like than in the HNC/HCN case, the tunneling correction is expected to be smaller, although this must be verified quantitatively.

Electronic-Structure Methods

For each reaction, all stationary points must be optimized and characterized at the following six levels of theory (work as a team and divide the jobs among yourselves):

Label	Functional	Basis set
M1	M06-2X	def2SVP
M2	M06-2X	def2TZVP
M3	ωB97X-D	def2SVP
M4	ωB97X-D	def2TZVP
M5	B3LYP-D3(BJ)	def2SVP
M6	B3LYP-D3(BJ)	def2TZVP

In Gaussian16, the corresponding route-section keywords may be written as:

$$\text{M06-2X} : \texttt{M062X}, \tag{28}$$

$$\omega\text{B97X-D} : \texttt{wB97XD}, \tag{29}$$

$$\text{B3LYP-D3(BJ)} : \texttt{B3LYP EmpiricalDispersion=GD3BJ}. \tag{30}$$

All calculations for a given reaction and a given method must be performed consistently. The reactant minimum, transition state, and product minimum must use the same functional, basis set, integration grid, SCF convergence settings, charge, and spin multiplicity.

Tunneling Corrections

Classical TST

The first calculation must be performed without any tunneling correction:

$$\kappa(T) = 1. \tag{31}$$

This gives the classical transition state theory rate constant,

$$k_{\mathrm{classical}}(T). \tag{32}$$

Wigner Correction

The Wigner correction is a simple analytical tunneling correction based on an inverted parabolic approximation to the barrier. It uses only the magnitude of the imaginary transition-state frequency. If the imaginary frequency is expressed as a wavenumber $\tilde{\nu}^{\ddagger}$ in cm^{-1}, the Wigner correction is

$$\kappa_{\mathrm{Wigner}}(T) = 1 + \frac{1}{24}\left(\frac{hc|\tilde{\nu}^{\ddagger}|}{k_{\mathrm{B}}T}\right)^2. \tag{33}$$

The Wigner-corrected rate constant is then

$$k_{\mathrm{Wigner}}(T) = \kappa_{\mathrm{Wigner}}(T)k_{\mathrm{classical}}(T). \tag{34}$$

Eckart Correction

The Eckart correction uses an analytical one-dimensional asymmetric barrier. Unlike the Wigner correction, the Eckart correction requires information about both the forward and reverse sides of the reaction:

1. the reactant energy and thermochemistry;
2. the transition-state energy and imaginary frequency;
3. the product energy and thermochemistry.

The Eckart-corrected rate constant is

$$k_{\mathrm{Eckart}}(T) = \kappa_{\mathrm{Eckart}}(T)k_{\mathrm{classical}}(T). \tag{35}$$

For asymmetric reactions, such as HNC $\rightarrow$ HCN, the Eckart correction is usually more physically informative than the Wigner correction because it includes the reaction energy and the reverse barrier height.

Temperature Range

For each reaction and each level of theory, compute the rate constants over the following temperature range:

$$T = 100, 150, 200, 250, 300, 400, 500, 750, 1000, 1250, 1500\ \mathrm{K}. \tag{36}$$

Alternatively, a regular temperature grid may be used:

$$T = 100\ \mathrm{K}, 150\ \mathrm{K}, 200\ \mathrm{K}, \ldots, 1500\ \mathrm{K}. \tag{37}$$

The low-temperature region is included because tunneling effects are usually most visible at low temperature. At high temperature, the tunneling correction is expected to become less important and $\kappa(T)$ should approach unity.

Arrhenius Analysis

For each reaction, method, and tunneling model, construct an Arrhenius plot:

$$\ln k(T) \quad \text{versus} \quad \frac{1}{T}. \tag{38}$$

Students should compare and comment on the plots for:

1. classical TST;
2. Wigner-corrected TST;
3. Eckart-corrected TST.

The low-temperature points may deviate strongly from linearity when tunneling is important.

Comparison Between the Two Reactions

Students should explicitly compare the HNC isomerization and the formic acid isomerization.

Expected Differences

For HNC $\rightarrow$ HCN, the reaction coordinate involves significant hydrogen displacement. Therefore, the imaginary frequency may be large in magnitude and the barrier may be relatively narrow. This reaction is expected to show a stronger tunneling correction, particularly at low temperature.

For cis-HCOOH $\rightarrow$ trans-HCOOH, the reaction coordinate is mainly a torsional rearrangement. Since the effective moving mass is larger and the barrier is usually broader, the tunneling correction is expected to be smaller.

Questions

Students should answer the following questions:

1. Which reaction has the larger classical activation free energy?
2. Which reaction has the larger imaginary transition-state frequency?
3. Which reaction shows the larger Wigner correction at 300 K?
4. Which reaction shows the larger Eckart correction at 300 K?
5. Does the Eckart correction differ significantly from the Wigner correction?
6. For which reaction is the Arrhenius plot more strongly curved at low temperature?
7. Does the basis set affect the tunneling correction mainly through the barrier height, the imaginary frequency, or both?
8. Are the qualitative conclusions the same for M06-2X, ωB97X-D, and B3LYP-D3(BJ)?

References

[1] H. Eyring, *The Activated Complex in Chemical Reactions*, Journal of Chemical Physics, **3**, 107–115 (1935).

[2] E. Dzib, J. L. Cabellos, F. Ortíz-Chi, S. Pan, A. Galano, and G. Merino, *Eyringpy: A program for computing rate constants in the gas phase and in solution*, International Journal of Quantum Chemistry, **119**, e25686 (2019).

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